Timothy Choi

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Experience

Dassault Systèmes — **Germany**
Software Engineer — *2024 May –*
- Working on the appearance core team to provide materials to countless apps such as CATIA and SIMULIA
- Creating debug tools to efficiently search for and modify values of materials

Altacomm Technologies — **Australia**
Software Engineer Intern — *2024 Jan – 2024 Mar*
- Building a full-stack contact us and media release page with Rust, Actix, and Astro
- Polishing up the company landing page by increasing the responsiveness and adding animations

Global Technology Integrator — **Hong Kong**
Network / System Engineer — *2021 Aug – 2021 Nov*
- Spearheading the automation and parallelization of our tasks
- Helping resolve our staff shortage by reducing the manpower and time required by ~200%
- Mentoring new staff, helping them understand the technology

Education

Technical University of Munich — **Germany**
Exchange

The University of Adelaide — **Australia**
Bachelor of Mathematical and Computer Sciences, GPA: 6.33/7.0

Projects

Block Model Compression Algorithm (`tinnamchoi.github.io/projects/bsoc`)
C++, Algorithm Design, OOP, Git, Agile
- Implementing octrees to compress 3D data in $O(N \log_8 N)$ time
- Employing agile methodologies for a group of 7
- Enforcing good Git practices such as using conventional commits

Binary Search on Color (`tinnamchoi.github.io/projects/bsoc`)
Algorithm Design, JavaScript, HTML, CSS, Web Development
- Using the binary search algorithm to find the color that is closest to the user's desired color
- An attempt to combat indecisiveness when it comes to choosing colors

Hare-Clark C++ (`tinnamchoi.github.io/projects/hcc`)
C++, CMake
- Seeing how the CS Club committee was struggling to debug the `hcvote` Python library, I created this C++ library as a sanity check
- In the process of creating this program, I discovered multiple flaws in the original library, and decided to release this as a better alternative

Malcolm bot (`in-progress`)
API, SQLite, Python
- Access the LeetCode, Codeforces and CLIST API to conveniently fetch corresponding information
- Store database of past solved problems per user to generate graphs and statistics

Volunteering

Adelaide Rover Team — **Australia**
Software Developer — *2023 Jul – 2024 Jan*
- Employing best practices for seamless coordination and optimal outcomes
- Consolidate constants for streamlined management and easy access

Skills & Achievements

Languages

Proficient: C++ — **Intermediate / Beginner**: Bash, C, CSS, HTML, JS, LaTeX, MATLAB, Python

Human: English (fluent), Cantonese (native), Mandarin (advanced), German (beginner)

Prizes

Subject prize: Information and Communication Technology 2021 — *Achiever of the best results amongst peers*

Scholarships

University of Adelaide: 30% Global Citizens Scholarship — *Best possible scholarship for international students*